

Gait Detection using Wrist-worn IMUs for Home Rehabilitation of Stroke Patients

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Abstract

Stroke is a leading cause of long-term disability, necessitating prolonged home rehabilitation where clinicians require quantitative insight into a patient's functional progress. While wrist-worn Inertial Measurement Units (IMUs) can track arm movement, gait must be filtered out to isolate true functional arm use. We propose a dual gait-detection algorithm approach tailored for stroke patients: a real-time, computationally light detector for immediate feedback as well as a robust offline model for end-of-day analysis of functional arm use. To evaluate the algorithms, we record data on four stroke patients, with varying levels of impairment and assistive walking devices, performing a mix of activities including gait and arm-intensive non-gait movements. For real-time applications, an adaptation of Kheirkhahan's method achieved an $F1 = 0.805$ in clinical evaluation among patients not using a walker, but failed to detect gait in patients who used one. For end-of-day analysis, we developed StrokeNet by adapting and retraining an existing machine learning model. This approach achieved an $F1 = 0.938$ in patients not using a walker, and remained robust when including patients using walkers ($F1 = 0.869$). These algorithms not only handle complex edge cases like the use of canes and periodic non-gait arm activities, but also outperform the legacy algorithm requiring a less-accessible leg-worn sensor in addition to the wrist sensors. These results demonstrate the feasibility of high-accuracy gait detection using only wrist-worn sensors, providing a cost-effective solution for monitoring home-based stroke rehabilitation.

have no quantifiable insight into the progress of home rehabilitation.

Recent years have seen the rise of technology for wrist-worn sensors and complementary gait detection algorithms, typically in the form of smartwatches [3]. While most of these are aimed at the general population, the ease of wear of wrist sensors could prove to be beneficial for use in home-rehabilitation of individuals recovering from stroke. To this end, the Towards@HomeRehab project is being developed. It aims to use wrist-worn IMU sensors for monitoring functional arm use of patients that conduct rehabilitation in free-living conditions at home. IMU sensors contain an accelerometer, magnetometer and gyroscope to measure acceleration, magnetic field and angular velocity, respectively, in 3 dimensions each (x,y,z), from which specific patterns and behaviours can be observed.

For this assessment, the clinician is interested in seeing the amount of arm use ('energy') as well as the ratio between the impaired and non-impaired arm use. In detecting the functional arm use, the non-functional use should be filtered out in order to not contaminate the energy computations with passive movements. For this goal, a gait algorithm is developed which can then be applied to exclude periods of walking from the arm use calculations. A previous edition of the rehabilitation project required a leg-worn sensor which was exclusively used to detect walking. Not only are leg-worn sensors less accessible and more invasive to the user, but they also increase the cost of the setup as an additional sensor is required for the sole purpose of detecting gait. Therefore, the clinician requires a real-time gait detection algorithm using solely wrist-worn sensors in order to provide direct feedback to the patient during use, as well as an end-of-day algorithm to perform analysis on the entire day of data to see the hourly and total arm use of the patient and whether their targets have been achieved.

This paper first explores the already existing datasets and algorithms for gait detection from IMU sensors in Section 2. It then describes how data was collected and how several selected algorithms are developed and implemented to detect walking, for both real-time and end-of-day analysis, in Section 3. The results are then presented in Section 4 and discussed in Section 5 before the final conclusion of this study are summarised in Section 6. The data and code availability are also provided in Section 8.

1 Introduction

Stroke remains one of the world's leading causes of disability, with rehabilitation playing a vital role in the patients' recovery of motor skill necessary for activities of daily life (ADL) [1][2]. However, due to prolonged duration of the recovery, along with the high costs of required therapy and the availability of needed facilities, it is largely beneficial to provide the rehabilitation for patients at home. This can be an issue as many struggle to keep the motivation to perform the exercises on their own and the clinicians

Table 1: Summary of gait detection algorithms.

Algorithm	Method type	Original sensor site	Target population	Key parameters / thresholds
ADEPT [4]	Signal processing	Wrist	General	Template correlation threshold; stride template shape
ElderNet [5]	Deep learning	Wrist	Older adults; impaired gait	Fine-tuned classification probability
Hickey [6]	Signal processing	Lower back	General	Mean & SD thresholds on normalised acceleration
Kheirkhahan [7]	Signal processing	Hip	General (older adults in original)	Activity count threshold per window; robust to in between subject variability
MacLean [8]	Signal processing	Lower back	General	Multiple amplitude/frequency thresholds
Keren [9]	Signal processing	Wrist	General	Peak detection thresholds; signal variability bounds; dominant frequency range
Strackiewicz [10]	Signal processing	Wrist / hip	General	Acceleration magnitude; step frequency, amplitude
MultiMobility wrist [11]	Signal processing	Wrist	General	Adaptations of Hickey, Kheirkhahan and MacLean methods with wrist worn sensors

2 Related Work

The recognition of Activities for Daily Living (ADL) for Humans is becoming an increasingly popular task in which studies investigate how to identify and distinguish different activities that are performed in daily life using data from sensors worn on the body. The focus of this paper is to determine gait, which falls under the umbrella of ADL, from wrist-worn IMUs, which are frequently used in ADL recognition. Gait detection algorithms are also utilised in many familiar devices, such as smartwatches and mobile phones. As a result, there exists a plethora of open-source gait detection algorithms and publicly available datasets. This section aims to highlight the relevant algorithms and datasets to be used and adapted for the aim of detecting gait using wrist-worn sensors in stroke patients.

2.1 Existing Algorithms

A broad search for open source gait-detection algorithms was conducted, looking at processing approaches for IMU data. Existing methods for detecting walking were short-listed and are summarised in Table 1.

Several other algorithms were found but were not readily available as they required special permissions/licences. Such include the Machine Learning (ML) analysis of ADLs in stroke patients [12] and a signal processing approach comparing the Upper Limb (UL) activity for non-disabled patients and patients with stroke [13]. Other methods reviewed include the algorithms for estimation of step length [14] and review of arm swing characteristics during walking for Parkinson’s patients [15]. Overall, it was found that all algorithms exclusively use the acceleration data in order to determine gait.

2.2 Public Datasets

Due to the constraints of ‘gait detection’ and ‘wrist-worn IMUs’, the search for publicly available datasets was

reduced to datasets in which ‘walking’ (or similar) was listed as one of the performed activities, where IMUs or accelerometers were a sensor modality utilised and at least one was placed on the wrist. There was also a preference for test subject cohorts of individuals with impaired gait, for example the elderly, or those diagnosed with a condition that affects gait such as Parkinson’s Disease (PD) or similar. Furthermore, it was necessary for the datasets to be accurately annotated in order to compare predictions to ground truths. These datasets can be then used to evaluate/validate gait detection algorithms (see Section 4).

In total, over 20 public datasets for ADL recognition were found, of which four were saved for subsequent algorithm evaluation. Many were removed due to the fact that some datasets were not easily accessible and required a special license or because the data was not easily parse-able (see Section 9.1). The following datasets remained:

- **WearGait:** (100 Hz) Recordings of 33 individuals with Parkinson’s Disease (PD) and 19 age-matched controls synchronized to a gait reference system. Frame-by-frame annotation of participant actions during gait and balance tasks was incorporated into the dataset using video cameras synchronized to all systems, wearing multiple sensors including wrist-worn IMUs [16]. The participants performed walking trials where 6 activities are recorded (sitting, standing, walking, opening door, turning) for a total of roughly 3 minutes each.
- **WISDM:** (20 Hz) This dataset includes activity data from 51 healthy participants from an activity recognition project. Each participant performed each of the 18 activities for 3 minutes with a smartphone and

smartwatch (accelerometer and gyroscope each) [17].

- **HMP:** (32 Hz) Recordings of 16 healthy volunteers performing 14 Activities of Daily Living (ADL) while carrying a single wrist-worn tri-axial accelerometer [18]. Each activity spans from a few seconds to a few minutes.
- **BioCliTe:** (50 Hz) The dataset collected for this study consists of triaxial acceleration and triaxial gyroscope data from 20 healthy subjects performing a set of free-living activities (7 ADLs) for a total of roughly 5 minutes while wearing a smartwatch on their preferred wrist [19]

3 Methodology

This section provides an overview of the data collection procedure used to obtain data to train and evaluate the algorithms on as well as a description of the gait detection algorithms which are further examined and adjusted.

3.1 Data Collection

The sensors used for the Toward@HomeRehab project are taken from a developing edition of the QSense Motion Platform, pictured in Figure 1. The system consists of several IMU sensors, which record data at a 50Hz sampling frequency, along with the central processing hub, which is equipped with its own IMU sensor as well as a processing unit to store data from all sensors and run a real-time gait detection algorithm. The system is initially configured using the QSenseHub application with the hub connected to the computer in docked mode. The recording of the data happens in wireless mode, during which the data from the sensors is transmitted to the hub where it is stored and can be processed (in limited capacity) if necessary. After the recording, the hub can again be connected in the docked mode allowing the data to be offloaded for further processing.



Figure 1: QSense Motion Platform [20]

3.1.1 Self-recorded Datasets

To evaluate algorithmic performance and the feasibility of the QSense sensor setup, data was recorded from both healthy subjects and stroke patients performing a variety of tasks. In total, four datasets were recorded and are listed below:

- **Baseline:** Self-recorded data of authors performing 4 basic activities in isolation (walking, standing, stairs and sitting) for roughly 3 minutes each.
- **Clinical:** Data recorded on 5 stroke patients performing several walking trails with varying intermittent non-gait activities for a total of roughly 10 min-

utes each. Non-gait activities involved: folding laundry, placing puzzle pieces and dominoes, watering plants, washing hands, reading a magazine, and exercising using an arm cycle machine. The leg sensor was applied in these recordings to enable comparison of wrist performance to leg classification performance.

- **Edge Cases:** Self-recorded data of authors performing 10 edge cases of walking (stairs with rail, hands in pockets, holding phone, limping, simulating walk of stroke patient, crutches, cane, walker, vacuuming) in isolation for approximately 3 minutes each as well as one longer recording of free-living with no walking.
- **Multiple Activities:** Self-recorded data of authors performing multiple activities in one recording (combination of walking, standing, sitting, eating, vacuuming, reading, washing hands, simulating walk of stroke patient, varied gait speed) for a total of roughly 10 minutes each. The leg sensor was applied in some of these recordings in order to compare wrist performance to leg classification performance.

Lastly, we also annotated walking using binary labels and utilised a dataset recorded by a colleague at Erasmus MC, in which stroke patients were performing semi-structured trials for ADLs in the kitchen. The focus of this recording was not walking so minimal walking periods occur, and none longer than 15 seconds. These recordings will be referred to as *Free Living*. They were also recorded at 50Hz with the same setup (QSense Motion Platform) used. The annotations were made using the video recordings which were synced with the IMU data.

In all cases, the self-recorded data is examined to remove any faulty timestamps or to segment it between the periods of the missing data. This is done to ensure data continuity in the input of the algorithms.

3.2 Algorithms

In Section 2.1 a shortlist of existing algorithms was established. We attempted implementations of all algorithms to obtain an initial indication of performance and usability. Any algorithms that could not be implemented or produced poor results on the *Baseline* dataset were discarded. The remaining algorithms are explored further and discussed in this section.

3.2.1 ElderNet

ElderNet is a deep learning approach to gait detection using wrist-worn sensors, optimised for the elderly and those with impaired gait. ElderNet uses 3-axis acceleration data in order to obtain a probability of gait based on a window of data [5].

ElderNet’s model development pipeline can be seen in Figure 2. ElderNet comprises a frozen pre-trained feature extractor, a ResNet-V2 (18 layers) from UK Biobank Self Supervised Learning (SSL), producing 1024-dimensional features from 10-second input windows, paired with a customisable head and task-specific classifier. The SSL model was modified to include the large unlabelled dataset named RUSH Memory and Aging Project (MAP), which includes wrist-worn sensor data from over 1000 elderly with mixed gait abilities for 10 days. The model was then fine-tuned on the labelled Mobilise-D technical vali-

dation study, consisting of 83 adults wearing a wrist-worn accelerometer for approximately 2.5 hours. The dataset consists of adult with varying health conditions, including Parkinson’s Disease, proximal femoral fracture, chronic obstructive pulmonary disease, congestive heart failure, as well as healthy adults [5].

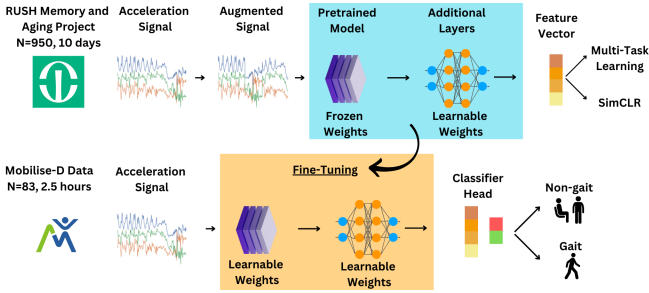


Figure 2: Model development pipeline for ElderNet [5].

This results in a model which takes accelerometer windows of shape $[3 \times 300]$ (10s at 30 Hz following the sampling frequency of the training data) in mini-batches. This is inputted into the 1D Convolutional Neural Network (CNN) backbone followed by a fully connected head. The CNN backbone processes 3-axis acceleration data in five layers, which double in channels while downsampling by half, resulting in a 1024 dimensional feature map for 9 time steps at the final layer. Each layer applies 1D convolutions, residual blocks, batch normalization, ReLU activation, and anti-aliased downsampling to reduce temporal resolution while increasing feature dimensionality [5]. The early layers learn low-level features such as peaks and troughs per axis while the later layers learn complex high-level abstractions from all axes. Following the backbone, a projection head comprising three fully connected layers ($1024 \rightarrow 512 \rightarrow 256 \rightarrow 50$, with ReLU non-linearities) produces a low-dimensional representation. This representation is then fed to a binary classifier that outputs two logits (gait vs non-gait) [5]. The gait probability is obtained by applying softmax to these logits and selecting the gait class probability, which maps the outputs to normalised class probabilities that sum to 1.

In order to run inference on data, we resample the input data to 30 Hz and extract 10s windows from the data to match the ElderNet input. This windowing is repeated every second to obtain an inference for the previous 10s of data. A window is considered gait if the rounded probability is equal to 1, essentially a confidence threshold of 0.5.

3.2.2 StrokeNet

StrokeNet follows on from ElderNet by slightly modifying the architecture in order to facilitate inference on smaller windows and retraining on self-collected data using the QSense IMU sensors.

Architecture

We desired analysis on smaller window samples in order to detect shorter walking periods (< 10 s). We chose for 2s windows in order to detect short walking periods whilst maintaining sufficient samples to perform analysis.

At 50Hz sampling frequency, this gives us 100 samples per window. However, at 100 samples this results in a sequence length of size 3 for the 5th downsample layer in the backbone, which is inoperable for a kernel of size 5. Therefore, the last downsample layer was removed to maintain a sample size of 6, larger than the kernel size. Finally, the circular padding in the backbone was replaced by zero padding as to not make assumptions about the periodicity of the signal for such short windows.

Retraining

Given that ElderNet is fine-tuned for a different target demographic and on data using different sensors, we retrain the model on labelled data we collected using the QSense sensors and annotated ourselves, namely the *Edge Cases*, *Mixed Activities* datasets. The *Free Living* dataset is also included to capture stroke-specific characteristics which impact the acceleration patterns during gait.

We extract windows of acceleration data of length 2s with 1s stride and split the data using a subject-aware 80/20 training/validation split. Therefore, no subject-activity IDs appear in both sets. The training uses the validation F1 score as the metric to optimise and is configured to use the Adam Optimiser with differential learning rates (Backbone: $1e-5$ [slower/fine-tuning], Head: $1e-4$ [faster, task-specific]), a weighted Cross Entropy Loss to handle the class imbalance, a learning rate scheduler that automatically reduces the learning rate when the training plateaus (waits for 5 epochs of no improvement before halving learning rate) for 40 epochs of batch size 64. Early stopping was manually implemented after we noticed that the performance did not increase for 9 consecutive epochs, at epoch 28. The retraining produced a new set of weights which could be loaded with modified architecture of ElderNet, which we hereinafter refer to as StrokeNet.

3.2.3 Strackiewicz

Strackiewicz et al. developed a "one-size fits most" gait detector for smartphones, smartwatches, and wearable accelerometers which measure tri-axial acceleration data [10]. Walking is classified based on three inherent features of walking: periodicity, intensity and duration [10].

Original Strackiewicz Method

The Strackiewicz method is a batch processor, which uses a Continuous Wavelet Transform (CWT) to decompose the acceleration signals, capturing the transient, rhythmic nature of steps without assuming stationarity. It continues by computing the peak-to-peak amplitude per second to remove any seconds where the amplitude is below a preset amplitude threshold, to avoid further analysis on sedentary periods. It then applies a context-aware three-zone frequency discriminator based on the step frequency. A final requirement for temporal continuity is implemented to ensure the dominant frequency is present in at least T seconds and does not drift more than β , to convey the fact that walking is a sustained, rhythmically stable activity [10].

Due to difficulties in applying this algorithm locally, the method was adapted slightly to enable its functionality, which is discussed below.

Adaptation

This method was adapted to produce a simplified, real-time gait detector using a triple logic gate which hereinafter will be named as 'Strackiewicz A'. Firstly, we implement a Fast Fourier Transform (FFT) instead of the CWT on a single windowed segment to determine the global power spectrum for that window. This is because the CWT and the dominant peaks analysis require the entire signal and is computationally very expensive. The dominant frequency for this window is computed and used in the first logic gate, where the frequency must fall between the preset minimum and maximum step frequencies. The power of this frequency, computed as the magnitude squared, acts as the second logic gate and must fall within the preset power band. Finally, the amplitude is calculated as the standard deviation of the accelerations in the window, which acts as the third logic gate. If a window satisfies these three conditions, it is considered gait and outputs a 1, and if at least one of these conditions is violated, a 0 (non-gait) is outputted. We look at 2s windows with a stride of 1s, and incorporate the previous detection state with the current detection state to determine if the window is gait, by requiring both to predict gait, as a minimal temporal continuity constraint in order to mitigate false positives.

Threshold Optimisation

The adapted method uses minimum and maximum thresholds for three logic gates. In order to obtain optimal threshold values, we use a Bayesian Optimiser. Given the high dimensionality and wide range of typical values for this search, a Bayesian Optimiser can navigate this wide search field by exploiting promising directions for. We leverage the *Multiple Activities* dataset in a 70/30 training/test split for the optimiser, where it maximises the F1 score.

3.2.4 Hickey Adaptation

The original Hickey method was developed for macro-gait analysis of healthy subjects in free-living conditions [6]. This was done by processing the accelerometer data collected with IMU sensors on the lower back with the sampling frequency of 100Hz. This approach was validated for prolonged uncontrolled free-living environments by recording 10 healthy participants for 1 hour on two separate occasions with the camera recording used for annotations.

The classification is done with window-based calculations of the acceleration signal variability and orientation thresholds. The data is filtered, with the offset removed, and further separated into windows of 0.1s with each window's mean and standard deviation calculated. The main classification step for each window occurs for the mean below a defined threshold, T_{up} , to signify the upright position of the user, and the standard deviation above a different threshold value, T_{still} , to ensure a certain level of range of motion of the movement. Once the window is labelled with gait presence, continuous bouts of 2.25s are merged and the bouts smaller than 0.5s are discarded.

In adapting to the wrist worn position, the gravitational component is removed before the acceleration norm is taken, and the two threshold values are retuned [11]. The gravity removal is done by applying Butterworth low pass

filter (with a cut-off frequency of 0.25Hz) on the data to separate the gravitational influence on the values, which is then subtracted from the raw data. In changing the threshold values, the basic principle for the standard deviation assessment remains that the movement is required to be sufficiently noisy (higher deviation) compared to the still periods, with the threshold value now set to 0.98 (tuned at 0.1 with acceleration in G units). The evaluation of the mean, which was originally used as the upright threshold for lower back position is now instead seen as a maximum activity threshold, with the value set to 9.5 [21].

Threshold Optimization

We further tune this approach on the *Edge Cases* dataset by prioritizing the performance on the edge cases with asymmetric gait patterns (when having a limb or when using a cane), with the goal of maximizing the F1 scores for those conditions. The final overview of the process can be seen in Figure 5 in the Appendix (Section 9.2).

3.2.5 Kheirhahan Adaptation

Kheirhahan created a signal processing method for adaptive walk detection by evaluating the activity counts [7] from the accelerometer data. This approach was originally developed for hip-worn sensor position, and validated on the data from 22 healthy participants when performing leisurely walking, rapidly walking and stationary tasks. The main strength of this algorithm is that it creates a detection method adaptive to a small window of activity count data (10-15 seconds).

The algorithm again relies only on the acceleration data. The activity count per 1s bins is computed directly from the triaxial acceleration. The process of calculating the activity count is taken from the ActiLife software (ActiGraph™, Pensacola, FL, USA). The relative activity per window is compared against a provided threshold.

The calculation of the activity counts involves the down-sampling of the data with applying an anti-aliasing (Butterworth) bandpass filter. The ActiGraph specific filter is taken from the implementation which reproduced the official version with a custom band-pass filter [22]. The data is further downsampled with anti-aliasing filter again and truncated. It is then separated into the 1s bin based on its 8-bit representation.

This method then uses a sliding window approach to evaluate the activity count in each of the windows. The window step is set to 1s and the best performing window size is found to be 12s. The inactivity parameter per window is found from the relative activity count, found as the ratio of the activity in the window to the higher activity value in that window, with the outliers removed. This is done to ensure the method's robustness in performance in between varying subjects and adaptivity to a small window of activity count data[7]. The classification of gait is then assigned based on the activity threshold, T_{act} .

Adaptation for Wrist-worn Position

This method is adapted to the wrist position by taking the acceleration norm before the activity count is computer and by recalibrating the value for the window size, and activity threshold [11]. For this implementation, the

micro-walking bouts are also merged into the continuous walking about that are longer than 3s. The adapted version of the algorithm can be seen in Figure 6 in Section 9.2.

Tuning and Implementing the Standard Deviation Threshold

The values for T_{act} , the truncation range, R , the window size and step, and the outlier range were tuned to improve the overall performance of the self-recorded data (the data used in tuning is highlight in the Section 4). This was done by aiming to increase the F1 score, and prioritizing improving the performance for edge cases of asymmetric walking patterns (of using the cane/limping/simulating stroke).

When tuning the algorithm, we inspect the performance on several of the self-recorded tests where it was seen that the algorithm at times mistakenly identified a period as walking even when the range of motion was low. For this reason, we further implement a threshold check for the standard deviation, same as shown in the Hickey implementation using T_{still} , for each of the windows. This then discards the periods of insignificant motion and increases the precision of the algorithm. We further tune the value for this threshold with self-recorded data. The data from *Multiple Activities* dataset is prioritized in increasing the performance, furthermore we prioritize the increase in precision to the increase in the recall. The overview of the modified version is presented in Figure 7 in Section 9.2. The performance without and with the added threshold value is further evaluated in Table 2, referred to as 'Kheirkhahan' and 'Kheirkhahan mod.' methods respectively.

Real-time simulation

As the final goal of the application for home rehabilitation requires direct feedback, the real-time analysis of the gait is required. The adaptation of Kheirkhahan is chosen for this as it operated on relatively small window size and should not require the entire duration of the trial as compared to the other signal processing methods. In Section 4, this method will be further referred to as *Realtime*.

To evaluate this performance, we adjust the method to only "feed" a small window into the algorithm at a time, to simulate the data coming in. For this purpose, 13s windows are used, as the first and the last 2s of data will be used as padding, meaning that the prediction taken accounts only for the 9s window in the middle.

It is important to note that when the algorithm makes the predictions for each new window, which for the most part overlap with the previous one, if the window is labelled as walking, the previous prediction is overwritten, and it is not just assigned the last one second being labelled as walking. This is done to ensure consistency with the way the original algorithm is applied where any 1s bin which is labelled as walking at least once keeps that label.

4 Results

In this section, the algorithms discussed in Section 3 are evaluated using the public datasets found in Section 2.1 and the self-recorded datasets in Section 3.1.1. Further in-

depth results are presented for the *Clinical* dataset, which is to be interpreted as the final evaluation dataset. Finally, the results for the different wrist fusion strategies are shown. Throughout this section, the key performance metric of gait detection is the F1 score, which gives the best understanding of the efficacy of a detector as it balances precision and recall. The formulas of these metrics are shown in Equation 1, where TP, FP and FN represent True Positive, False Positive and False Negative predictions, respectively.

$$\begin{aligned} \text{Precision} &= \frac{TP}{TP + FP}, & \text{Recall} &= \frac{TP}{TP + FN} \\ F_1 &= \frac{2 \cdot \text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}} \end{aligned} \quad (1)$$

4.1 Cross-dataset Global Evaluation

We evaluate each of the methods mentioned in Section 3 against four labelled public datasets of wrist-worn accelerometers, described in Section 2.2, as well as our self-recorded and self-annotated data, described in Section 3.1.1. The results for each method on each dataset are presented in Table 2, where the global F1 score is given for each dataset, by concatenating the individual predictions made from each wrist to obtain a net performance score. The stated thresholds for signal processing algorithms (Strackiewicz A, Hickey, Kheirkhahan, Real-time) are the result of their respective optimisation procedures described in Section 3. In tuning the Kheirkhahan method, the window size, the window step and the outlier range performed the best with the default values of 9s, 1s and [20,90]% respectively. It is important to note that StrokeNet was trained on the majority of data in *Edge Cases*, *Multiple Activities* and *Free Living*, hence the scores given are from the validation subset.

4.1.1 Per-activity Breakdown

We further investigate the cross-dataset evaluation by looking at the per-activity performance of each method on the public datasets with labelled activities to discover its gait detection performance as well as its performance on other activities which could 'fool' the detector. Given that we are detecting walking purely from wrist movements, periodic arm movements from other activities may well be interpreted as gait. The full per-activity per dataset breakdowns are shown in Section 9.3 containing Table 7, Table 8, Table 9, Table 10. Non-gait activity annotations for the self-recorded data are not available as we only labelled walking periods in these datasets for simplicity.

4.1.2 Detailed Results of Clinical Dataset

Clinical is the most clinically relevant dataset for evaluating the performance of the detectors. The subjects align with the target group demographic for this rehabilitation device and each recording is done with same sensors as those will be used for the final application. Furthermore, we do not use any of this data to train or fine-tune our models meaning evaluation on this dataset gives us the best indication of the generalisation of our algorithms to new subjects.

The dataset initially contained 5 patients, although we removed Sub 2 from the analysis as this patient did not meet the target group demographic. A table displaying

Table 2: Global F1 performance comparison across datasets for all methods and at given configurations. Global F1 is calculated by examining both wrist predictions and merging results into one global score.

Dataset	Stracz. A	ElderNet	StrokeNet	Hickey	Kheirkh.	Kheirkh. mod.	Realtime ³
Thresholds	$F = [0.0643, 5.71]$ $P = [2.17, 234]$ $A = [0.00517, 1.08]$	Prob = 0.5	Prob = 0.5	$T_{up} = 9.5;$ $T_{still} = 0.05$	$T_{act} = 0.62;$ $R = [0.06, 1.2]$	$T_{act} = 0.62;$ $T_{still} = 0.1$	$T_{act} = 0.62;$ $T_{still} = 0.1;$ Buffer = 2s
Baseline	0.930	0.625	0.971	0.837 ²	0.985	0.993	1.000
Clinical	0.567	0.622	0.839	0.566	0.565	0.566	0.575
Edge Cases	0.819	0.346	0.968 ¹	0.557 ²	0.881 ²	0.861 ²	0.845
Multiple Activities	0.892 ²	0.668	0.957 ¹	0.785	0.894	0.898 ²	0.891
Free Living	0.258	0.030	0.360 ¹	0.220	0.161 ²	0.130	0.158
WearGait	0.797	0.330	0.843	0.847	0.848	0.848	0.902
HMP	0.772	0.552	0.650	0.509	0.669	0.669	0.681
BIOCLITE	0.483	0.294	0.366	0.327	0.567	0.568	0.541
WISDM	0.619	0.136	0.402	0.293	0.568	0.572	0.570

¹ Validation subset only

² Dataset used for threshold optimisation of algorithm

³ Realtime implementation of Kheirkhahan

Table 3: Per-subject global F1 performance on *Clinical*

Subject	Aid	Leg Class. ¹	Stracz. A	ElderNet	StrokeNet	Hickey	Kheirkh.	Kheirkh. mod.	Realtime ²
Sub 1	Walker	0.762	0.048	0.561	0.743	0.536	0.023	0.023	0.041
Sub 3	—	0.754	0.723	0.655	0.915	0.596	0.734	0.734	0.719
Sub 4	Cane	0.836	0.803	0.688	0.904	0.659	0.724	0.727	0.744
Sub 5	Cane	0.896	0.717	0.728	0.857	0.509	0.753	0.754	0.737

¹ Legacy QSense gait detection algorithm using additional IMU on the thigh

² Realtime implementation of Kheirkhahan

detailed results for each patient can be seen in Table 3.

Table 3 also includes the performance of the leg classification algorithm, developed by QSense to detect gait from a sensor worn on the thigh. This algorithm achieved a global F1 of 0.798 for this dataset.

Due to ElderNet’s poor performance on the majority of datasets including this one, it will be excluded from further results.

4.2 Optimal Wrist Fusion Strategy

Up to this point the global F1 scores are calculated by taking all the data from both wrists individually before concatenating their results to obtain the net performance. One could, however, determine walking by using the predictions from only one wrist or a fusion of the two. We investigated different wrist fusion strategies using the two best performing algorithms, namely StrokeNet and Kheirkhahan real-time method, on the *Clinical* dataset. Due to the presence of faulty data, all fusion techniques fall back to classifications based on the data from one wrist if the data from the second one is faulty or not available.

4.2.1 StrokeNet

Several approaches were considered for StrokeNet. The wrist fusion strategies are as follows:

- **Affected:** Use only the affected/impaired wrist prediction
- **Unaffected:** Use only unaffected/unimpaired wrist prediction
- **ACC Average:** Average of the both wrist accelerations to compute probability of gait

- **Probability Average:** Average of both wrist probabilities

- **Voting AND:** Both wrists must predict gait

- **Voting OR:** Only one wrist (or both) must predict gait

ROC Curves for the performance of StrokeNet on the *Clinical* are generated and shown in Figure 3. The best F1 thresholds and their respective thresholds are also calculated for each fusion strategy and are in Table 4. The best performing strategy is the Probability Average which achieves $F1 = 0.869$ at a confidence threshold of 0.500. This is an increase of 3.6% relative to the global F1 using all wrist predictions. When excluding Sub 1 who uses a walker, Probability Average fusion achieves an $F1 = 0.938$. The threshold of 0.5 is almost identical for all subjects.

4.2.2 Kheirkhahan real-time

Two fusion strategies for merging the data from the two wrists are tested for the real-time simulation of the Kheirkhahan modified method. We will compare them against single wrist performance from either arm:

- **Affected:** performance from the single wrist, looking at the affected/impaired side.
- **Unaffected:** performance from the single wrist, looking at the unaffected/impaired side.
- **Fusion activity:** performance from both wrists, where the combined relative activity count per window is taken.
- **Fusion classifier (AND):** performance from both

Table 4: Per-subject best F1 scores with optimal thresholds for each wrist fusion strategy using StrokeNet on the *Clinical* dataset. Global row shows performance aggregated across all subjects.

Subject	Affected		Unaffected		ACC averaged		Prob. averaged		Voting (AND)		Voting (OR)	
	F1	Thresh.	F1	Thresh.	F1	Thresh.	F1	Thresh.	F1	Thresh.	F1	Thresh.
Sub 1	0.721	0.017	0.779	0.991	0.714	0.022	0.771	0.495	0.741	0.016	0.763	0.991
Sub 3	0.916	0.649	0.925	0.904	0.897	0.090	0.940	0.500	0.941	0.002	0.935	0.947
Sub 4	0.920	0.868	0.925	0.980	0.669	3.6×10^{-9}	0.942	0.500	0.942	0.484	0.926	1.000
Sub 5	0.916	0.983	0.813	0.342	0.533	3.9×10^{-8}	0.938	0.500	0.884	0.001	0.912	0.991
Global	0.839	0.868	0.842	0.947	0.633	2.1×10^{-5}	0.869	0.500	0.849	0.016	0.862	0.991

Table 5: Per-subject F1 performance of different fusion techniques with realtime Kheirkhahan

Subject	Affected	Unaffected.	Fused activity	Fused classifier AND
Sub 1	0.031	0.052	0.044	0.000
Sub 3	0.698	0.743	0.702	0.732
Sub 4	0.757	0.734	0.750	0.764
Sub 5	0.941	0.608	0.654	0.942

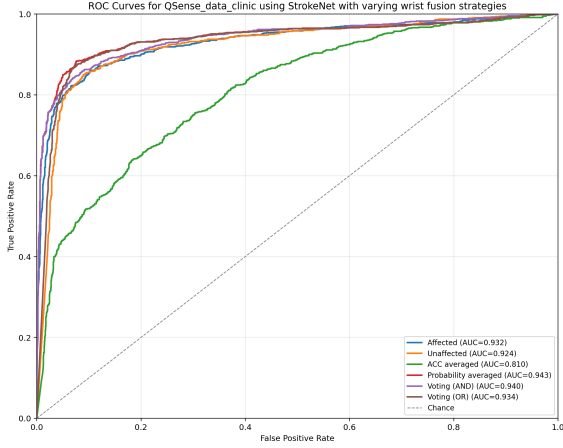


Figure 3: ROC Curves for different wrist fusion strategies performed with StrokeNet on the *Clinical* dataset with Area Under Curve (AUC) scores

wrists, where walking is detected only if both wrist classify the window as walking.

Fusing the data at the classifier with either wrist predicting walking (OR) is not considered due to its deteriorating effect on the precision.

The F1 scores of these approaches are summarized in Table 5. The results show that fusion at the classifier level further improves the performance of the algorithm. The weighted average F1 score for this approach is found to be 0.595 when including patient 1 and 0.805 when excluding patient 1 who uses a walker. In comparison, the weighed average F1 score when using the approach per wrist is found to be 0.566 when including and 0.733 when excluding patient 1.

5 Discussion

The results in Section 4 give a top-level overview of the performance of each algorithm. In this section, we analyse the results more thoroughly to pick up on the relative strengths and weaknesses of the algorithms. This section primarily focusses on the best performing algorithms: StrokeNet, Straczkiwicz A., Kheirkhahan modi-

fication and its realtime implementation.

5.1 Cross-dataset Analysis

The cross-dataset global performance evaluation enables us to investigate if the algorithms generalise well to unseen data and a wide range of subjects and activities.

Special consideration should be given to the *WearGait* dataset, as its population (PD/elderly) closely matches the intended target demographic for these algorithms, whilst *WISDM*, *HMP* and *BioCliTe* contain healthy and often young subjects. Healthy and young subjects may perform activities differently as to how stroke patients would due to differing levels of mobility, leading to contradictory wrist acceleration patterns between the two populations. Furthermore, *WISDM* and *HMP* use a sampling frequency of less than 50 Hz, meaning the data has to be upsampled to 50 Hz first before it can be analysed using StrokeNet, which can lead to interpolation errors. We see that the Hickey adaptation approach also cannot perform well on low sampling frequencies as the core implementation requires the sampling frequency above 34Hz and therefore cannot perform well on *WISDM* and *HMP* datasets without adjustment. In addition, *HMP* is evaluated by concatenating multiple recordings of one subject performing different activities, some of which only last a few seconds, which does not accurately portray real-life unstructured living. Finally, the *Free Living* dataset contains very short walking bouts ($\approx 3s$) so it is also difficult to detect walking on such short periods.

When examining the per-activity breakdown in Table 7, Table 8, Table 9, Table 10, we look at the accuracy as the main criterion for non-gait activities since the precision, recall, and F1 score cannot be calculated as no true positives are present. The Hickey adaptation is excluded from Table 7 and Table 9 as the adjustment of the algorithm to the lower frequency deteriorated the performance. We see that almost all methods have difficulties distinguishing certain activities involving hand movements from gait. Only ElderNet accurately detects most of these this as non-gait ($Acc > 0.9$) but when examining the results with more a closer ete, we see that the recall is extremely low

during gait activities ($R < 0.2$) implying that the algorithm is almost never predicting gait, even when gait occurs, leading to a high accuracy in non-gait activities. This is also reflected in the low F1 scores ($F1 < 0.5$). The poor ElderNet performance also occurred when varying the confidence threshold.

The low accuracy achieved by the algorithms for certain arm-intensive activities is to be expected given that movements like dribbling a basketball, combing the hair and brushing teeth induce similar rhythmic arm patterns as walking does. However, these false positives are not considered to be a major issue as the target group demographic is very unlikely to play sports nor would these activities occur for extended periods throughout the day, significantly impacting the patient’s energy calculations.

Furthermore, there are also varying performances for the same activity and detection algorithm between datasets. For example, StrokeNet achieves an accuracy of 0.943 and 0.944 for eating meat and eating soup, respectively, in the *HMP* dataset, but scores an accuracy of 0.409 and 0.525 for eating pasta and eating a sandwich. This is likely due to differences between subjects as well as differences in the test and recording setup, where different sensors of different sampling frequencies are used as well as different annotation techniques.

Analysis on the public datasets paints a picture that the algorithms can be misled by certain activities involving repeated arm movements. We further investigate this in our self-recorded data *Multiple Activities* by performing/simulating a range of activities in between walking periods as well as a variety of gait types (see Section 3.1.1). The high F1 scores ($F1 > 0.89$) show that almost all algorithms were able to distinguish these arm-intensive non-gait activities from gait. However, these results must be interpreted with caution as many of the best performing algorithms were either trained or fine-tuned on this dataset.

It can be seen that most algorithms detect gait whenever it occurs, with consistently high F1 scores for gait activities across all datasets, especially for StrokeNet and Kheirkhahan. Overall, this shows that these algorithms detect walking very well, but can be misled by activities which involved periodic arm movements.

5.2 Clinical Analysis

Despite its limited size, the *Clinical* dataset offers the most representative evaluation of algorithm performance, as it reflects clinically relevant, previously unseen data involving a range of activities. For this analysis, we limit this discussion to the best performing algorithms: StrokeNet, Strackiewicz A and Kheirkhahan.

5.2.1 Walking Aid

It is very important to note the walking aid used by the patients, indicated by the ‘Aid’ column in Table 3. Clearly, using a walker has a detrimental impact on Signal Processing based algorithms like Kheirkhahan and Strackiewicz A. This likely reflects the absence of arm swing, resulting in higher-frequency, lower-amplitude signals, driven by walker-induced vibrations or natural arm tremor. Example of this can be seen in Section 9.4 in Figure 8, where the figure shows the more pronounced walking patterns

for patient 3, who did not use a walker, whereas patient 1 did. However, these stable acceleration patterns also mean that the overall energy during these periods is not large, thus not accumulating large energy sums over time, and it could be deemed that the arms are now performing functional tasks given that they are supporting body weight. Therefore, it is not unreasonable to state that should these walking periods using a walker not be excluded from the energy analysis, the overall energy analysis would still remain accurate.

StrokeNet’s performance is also hindered by the use of a walker, but still remains within a clinically acceptable range for gait detection (global $F1 = 0.743$), showcasing that the machine learning model can be used to detect gait even when using a walker. We exposed the model to recordings of subjects using a walker so we can assume that the model has learnt the related acceleration patterns associated with walking with a walker.

Many stroke patients use a cane or a walking stick as a support aid when walking, which produces a slightly altered gait pattern in comparison to conventional walking, generally reduced arm swing and lower frequency steps. It is therefore important that the algorithm is robust also to the use of a cane when walking. The performance of StrokeNet, Strackiewicz A and Kheirkhahan realtime are not greatly hindered when walking with a cane, with strong global F1 results of above 0.857, 0.717 and 0.719, respectively. All three do not vary much from Sub 3 who walks without any aid.

5.2.2 Comparison with Leg Classification

Given that the leg gait pattern remains mostly unchanged when using a walking aid, it is interesting to compare the performance of the leg classification algorithm using a sensor worn on the thigh to the performance of the wrist-worn algorithms. Indeed, the leg classification achieves the highest global score for Sub 1 using a walker ($F1 = 0.762$), greatly outperforming the wrist-worn signal processing based algorithms. However, StrokeNet achieves a performance on par with the Leg Classification (global $F1 = 0.743$) and in fact outperforms the leg classification when fusing the wrist probabilities ($F1 = 0.771$). The leg classification continues to score well on the rest of the patients, however StrokeNet achieves a higher global F1 (0.869 [when fused] vs 0.798) implying that StrokeNet can better detect gait from wrist-worn sensors than the legacy QSense algorithm that utilises an additional sensor on the leg.

The same is not the case for the Kheirkhahan implementation and its derivatives (Kheir. mod and Realtime), where the F1 score is only sufficiently high ($F1 = 0.805$ with fusion) for the group without the walker and performs poorly on the patient with a walker ($F1 = 0.000$ with fusion). These scores however, are only taken from four tests and the approach should be examined further with more patients for a more thorough comparison.

5.2.3 Non-gait activities

In Section 5.1, we found that all algorithms were susceptible to false positive detections for activities that involve periodic hand movements when evaluated on the public datasets. We further investigate this in *Clinical*

trials by asking the patients to perform exercises involving hand/arm movements as mentioned in Section 3.1.1. StrokeNet was able to distinguish the gait periods from non-gait periods extremely well, especially for Subjects 3, 4 and 5, highlighting StrokeNet’s strong performance in detecting gait and ability to not be misled by strong periodic arm/hand movements whilst generalising to unseen, clinically relevant subjects.

On the other hand, the Straczekiewicz A algorithm often had many short false positive periods during washing hands and folding laundry activities, particularly for the patient’s non-affected side, given that this arm/hand is used more than the impaired arm when conducting these tasks. Furthermore, the algorithm identified the entire arm cycle machine use period as gait, further highlighting that strong periodic motions can mislead the detector.

The same issue is seen with the Kheirkhahan implementation, where the repetitive motions are identified as walking. Here the fusion approach at the classifier level improves this issue as it relies for both wrists to be seen as predicting walking. We seen that this improves the performance more significantly for patients with less active affected side as well, as the example of patient 5 show. Patient 5 was less mobile (compared to patient 4 using the same type of aid) and therefore relied more heavily on the non-affected side in the performance of the tasks, allowing for the affected side to be a better predictor of gait as the rhythmic motions were not as strongly pronounced.

5.3 Real-time vs End-of-day

A final consideration is the intended deployment of the algorithms. They may be used either for real-time onboard detection, providing direct sensor feedback, or for offline end-of-day analysis of patient IMU data by clinicians. A key aspect of a real-time detector is that the data is handled in a unidirectional buffer where it can only see current and previous timestamps with no access to future events. On the contrary, the end-of-day algorithm can make use of future events for a given timestamp which can provide benefits in the data processing e.g. standardisation based on global mean.

Furthermore, in order to be implemented as a real-time detector onboard the QSense hub, the algorithm must computationally light. The computational load characteristics of the best performing algorithms are found in Table 6. Discussions with team behind QSense platform found that the algorithm can have a maximum computational load of approximately 20kB in order to be mounted on the sensor, and that this load is traded-off with the power consumption.

Table 6: Computational characteristics of gait detection algorithms.

Algorithm	Memory ¹	Model Size ²
Kheir. fused	0.512 kB	—
StrokeNet	742.0 MB	42.57 MB

¹ RAM used during execution

² Memory required to store model weights (applicable only to machine learning models)

Table 6 shows that the Kheirkhahan fused signal pro-

cessing based algorithm is significantly lighter compared to the machine learning based StrokeNet. As a result, the signal processing methods like Kheirkhahan are much better suited to run onboard the QSense IMUs as a real-time gait detector, whilst the machine learning based StrokeNet is more appropriate as an end-of-day which the clinician can run on their local machine, due to its extremely large computational load.

5.4 Limitations & Recommendations

After reviewing the methodology and analysing the results, we identified several limitations and recommendations to address them.

5.4.1 Annotation and Free Living Data

The self-recorded data also required self-annotation. We employed a technique of noting the time of when an event/activity starts during recording to the nearest second. This limits the accuracy of our annotations which is especially significant for activities that last only a few seconds.

This is extremely relevant for the Free Living dataset, which was recorded by a colleague but not yet annotated for walking. Fortunately, the recordings were videotaped so we could retrospectively annotate the recordings using the second timestamps in the video. These second timestamps had to be manually synced with the timestamps of the sensors, which involved comparing the video footage to the acceleration patterns and identifying matching periods during the calibration phase of each recording. However, this gives rise to multiple sources of errors in addition to the fact that event timestamps cannot be recovered to the exact millisecond as it was only possible to sync the video timestamps to the sensor timestamps to the nearest 0.1s. Furthermore, it is difficult to discern when gait precisely stops and starts, especially for patients who walk very slowly or shuffle along the ground before walking. It should also be noted that these recordings contained very few walking periods, and present ones were relatively short (≈ 2 -3s, most under 10s). Thus, non-detection of these periods would not greatly skew the functional arm us energy analysis of the patient.

5.4.2 Insufficient Data

In addition, the *Clinical* dataset is too small to arrive at statistically significant outcomes regarding the efficacy of the gait detectors. Currently, the dataset gives a good indication of the expected performance of each algorithm once the device is deployed for home rehabilitation use, but in order to state the detection performance with high confidence, data from many more patients, who span the entire target group demographic, should be recorded whilst performing a wide variety of activities for an extended period of time (e.g. a whole day).

5.4.3 Incorporating Magnetometer and Gyroscope

All algorithms found were solely based on acceleration data in order to detect gait. However, the QSense IMUs include a magnetometer and gyroscope which measure the magnetic field and angular velocity, respectively. This could provide more stable and accurate motion tracking.

It is therefore recommended to investigate whether these measurements can be incorporated into the algorithms to improve their detection performance.

5.4.4 StrokeNet Improvements

With the addition of more clinically relevant data, it is also recommended to retrain StrokeNet again to further fine-tune the model to improve its performance on stroke patients. The magnetometer and gyroscope data could also be included in the retraining to exploit more data, potentially improving detection performance. Furthermore, it is also recommended to tune the hyperparameters of the model and its training, such as the window size and stride, optimiser, learning rates and scheduler and batch size.

5.4.5 Kheirkhahan Modification

The comparison of the results with and without the additional threshold check on the standard deviation are shown in Table 2. We see that the threshold improved the results to a certain degree for the *Clinical* dataset, as well as on the *Baseline* and *Multiple Activities*. This improvement is however not significant, with only 0.001 change on the *Clinical* set used as evaluation, and should be considered in terms of the computational power required for this step. Taking the standard deviation for 9s of data is a computationally expensive step and avoiding or adjusting it could be more beneficial for the application.

5.4.6 Limitation of Realtime Implementation

The greatest limitation of the real-time Kheirkhahan implementation is its inability to detect walking from patients using a walker. Therefore, periods of walker-use will be viewed as functional arm use instead of non-functional arm use. The scale of this impact can further be examined by recording more data with the patients using a walker and calculating the energy that is not filtered out by the algorithm.

A possible solution to this could also be looking directly at the activity count estimations which are computed in the Kheirkhahan method, and filtering out all the periods where the activity count is at zero. Further research should be done into seeing how the functional arm use, for movement such as reaching or lifting, are reflected on the energy estimate.

Furthermore, the Realtime Implementation has a tendency to predict false positives for movements that involve arm movements. As the primary candidate for real-time detection, this approach risks demotivating patients by underestimating their activity, as movements may be misclassified as walking and excluded from functional arm-use energy calculations. Further tuning on *Clinical* dataset as well as more clinically relevant data is recommended to mitigate any false positives.

5.4.7 Faulty Data

When recording data using the QSense IMUs, we also noticed periods of faulty data in some recordings, where repeated blocks of timestamps would overwrite the actual data. This could last for several seconds, minutes or for the rest of the recording. During data processing, we remove faulty data by filtering out repeating timestamps and flagging these periods but this could greatly affect the en-

ergy analysis the clinician performs at the end of the day, where potentially the entire day of data is invalid.

We investigated possible causes, such as the sensors being covered by clothing, being too close to or the line-of-sight being blocked by the torso. The only conclusion came from the hub being medially positioned on the upper arm (facing the torso) as that setup resulted in minors periods of faulty data. However, overall we were not able to find a clear setup or dependence between the position of the hub and the reliability for the data recording. It is therefore recommended to investigate this further.

6 Conclusion

The results indicate StrokeNet as a suitable algorithm in order to remove gait periods for the end-of-day energy analysis of the stroke patients' arm use. StrokeNet achieved the highest scores for the clinically relevant *Clinical* dataset (global F1 = 0.839, boosted to F1 = 0.869 when fusing the average of each wrist probabilities at a confidence threshold of 0.5 and further increases to F1 = 0.938 when patients who use a walker are excluded), where it was not misled by arm-intensive non-gait activities such as folding laundry, washing hands and using an Arm Cycle machine. It was even able to detect gait well for a patient using a walker (F1 = 0.771 when fusing probabilities), where typical gait patterns for the arm do not occur. Overall, it outperformed the Leg Classification algorithm which uses an additional sensor worn on the thigh (F1 = 0.798), despite arm gait patterns being greatly impacted by the spasticity that commonly results from a stroke as well as the use of walking aid, such as canes and walkers. StrokeNet achieved mixed results on public datasets, often incorrectly predicting gait for certain activities like brushing teeth and combing hair. It is therefore recommended to record and evaluate more data of stroke patients performing a wide range of activities and for an extended period of time. With additional data, it is also recommended to retrain StrokeNet to fine-tune its capabilities to detect gait for stroke patients. Due to the model's large computational requirements ($\simeq 740Mb$), it is unable to be implemented onboard the IMUs for real-time detection.

For the real-time implementation onboard the sensors giving real-time feedback to the patient, fused Kheirkhahan approach shows promising performance (F1 = 0.813). The algorithm is light in computation and does not require long periods of data input to classify walking. The main drawbacks of this approach are the inability to pick up walking activity from patients using a walker and the misclassification of tasks involving repetitive arm movements as walking. The first should be examined in the context of the goal of this application. The algorithm would label the periods when the walker is used as functional arm use. That should be evaluated by examining the energy used in those periods and comparing it to the energy from the daily arm use to assign how significant this issue is. The second could be addressed with the fusion technique using the data from both wrists, and should be further examined for a larger group of patients, spanning different activity levels.

7 Contributions

- **Hendrik Böckenhauer:** Implemented ElderNet, StrokeNet and Strackiewicz algorithms, annotated Free Living, developed end-of-day-analysis Jupyter Notebook for clinicians, and wrote sections Abstract, 3.2-3.2.1-3, 4.4.2.1, 5.5.2, 5.3, 5.4.1-4.
- **Tatiana Orlovskaja:** Implemented Hickey and Kheirkhahan, modified Kheirkhahan methods with the step form Hickey, implemented real-time simulation, annotated the recorded data, and wrote sections 1, 2-2.1, 3-3.1, 3.2.4-5, 4.2.2, 5.4.5-6.
- **Collaborated on:** making the recordings (both self-recorded and clinical), examining the faulty data and testing hub configurations, literature research, writing sections 2.2, 4.1, 5.1, 5.2.1-3, 5.4.7, 6, and making the company presentation.

8 Data & Code Availability

The data and code supporting this study are available on GitHub at: github.com/hbockenhauer/gait_detection. The company presentation summarising our work can also be found in the repository.

References

- [1] GBD 2023 Causes of Death Collaborators. “Global burden of 292 causes of death in 204 countries and territories and 660 subnational locations, 1990-2023: a systematic analysis for the Global Burden of Disease Study 2023”. In: *Lancet* (2025). PMID: 41092928; PMCID: PMC12535838. DOI: 10.1016/S0140-6736(25)01917-8.
- [2] Peter Langhorne, Fiona Coupar, and Alex Pollock. “Motor recovery after stroke: a systematic review”. In: *The Lancet Neurology* (2009). DOI: 10.1016/S1474-4422(09)70150-4.
- [3] Hari Prasanth et al. “Wearable Sensor-Based Real-Time Gait Detection: A Systematic Review”. In: *Sensors* 21.8 (2021). ISSN: 1424-8220. DOI: 10.3390/s21082727. URL: <https://www.mdpi.com/1424-8220/21/8/2727>.
- [4] Marta Karas et al. “Estimation of free-living walking cadence from wrist-worn sensor accelerometry data and its association with SF-36 quality of life scores”. In: *Physiological measurement* 42.6 (July 2021). DOI: 10.1088/1361-6579/ac067b. URL: <https://doi.org/10.1088/1361-6579/ac067b>.
- [5] Y. E. Brand et al. “Self-supervised learning of wrist-worn daily living accelerometer data improves the automated detection of gait in older adults”. In: *Scientific Reports* 14 (2024), p. 20854. DOI: 10.1038/s41598-024-71491-3. URL: <https://doi.org/10.1038/s41598-024-71491-3>.
- [6] Aodhán Hickey et al. “Detecting free-living steps and walking bouts: validating an algorithm for macro gait analysis”. In: *Physiological Measurement* 38.1 (Dec. 2016), N1. DOI: 10.1088/1361-6579/38/1/N1. URL: <https://doi.org/10.1088/1361-6579/38/1/N1>.
- [7] Martin Kheirkhahan et al. “Adaptive walk detection algorithm using activity counts”. In: *2017 IEEE EMBS International Conference on Biomedical Health Informatics (BHI)*. 2017, pp. 161-164. DOI: 10.1109/BHI.2017.7897230.
- [8] Mhairi K. MacLean et al. “Walking Bout Detection for People Living in Long Residential Care: A Computationally Efficient Algorithm for a 3-Axis Accelerometer on the Lower Back”. In: *Sensors* 23.21 (2023). ISSN: 1424-8220. DOI: 10.3390/s23218973. URL: <https://www.mdpi.com/1424-8220/23/21/8973>.
- [9] Karin Keren et al. “Quantification of Daily-Living Gait Quantity and Quality Using a Wrist-Worn Accelerometer in Huntington’s Disease”. In: *Frontiers in Neurology* Volume 12 - 2021 (2021). ISSN: 1664-2295. DOI: 10.3389/fneur.2021.719442. URL: <https://www.frontiersin.org/journals/neurology/articles/10.3389/fneur.2021.719442>.
- [10] Marcin Strackiewicz, Edward J. Huang, and Jukka-Pekka Onnela. “A “one-size-fits-most” walking recognition method for smartphones, smartwatches, and wearable accelerometers”. In: *npj Digital Medicine* 6.1 (2023), p. 29. DOI: 10.1038/s41746-022-00745-z. URL: <https://doi.org/10.1038/s41746-022-00745-z>.
- [11] Dimitrios Megaritis et al. “Real-World Wrist-Derived Digital Mobility Outcomes in People with Multiple Long-Term Conditions: A Comparison of Algorithms”. In: *Bioengineering* 12.10 (2025). ISSN: 2306-5354. DOI: 10.3390/bioengineering12101108. URL: <https://www.mdpi.com/2306-5354/12/10/1108>.
- [12] Pin-Wei Chen et al. “Measuring Activities of Daily Living in Stroke Patients with Motion Machine Learning Algorithms: A Pilot Study”. In: *International Journal of Environmental Research and Public Health* 18.4 (2021). ISSN: 1660-4601. DOI: 10.3390/ijerph18041634. URL: <https://www.mdpi.com/1660-4601/18/4/1634>.
- [13] Bailey Ryan R, Joseph W Klaesner, and Catherine E Lang. “Quantifying Real-World Upper-Limb Activity in Non-disabled Adults and Adults With Chronic Stroke.” In: *Neurorehabilitation and Neural Repair* (2015). PMID: 25896988; PMCID: PMC4615281. DOI: 10.1177/1545968315583720.
- [14] Seung Hyuck Shin and Chan Gook Park. “Adaptive step length estimation algorithm using optimal parameters and movement status awareness”. In: *Medical Engineering Physics* 33.9 (2011), pp. 1064-1071. ISSN: 1350-4533. DOI: <https://doi.org/10.1016/j.medengphy.2011.04.009>. URL: <https://www.sciencedirect.com/science/article/pii/S1350453311000907>.
- [15] Elke Warmerdam et al. “Quantification of Arm Swing during Walking in Healthy Adults and Parkinson’s Disease Patients: Wearable Sensor-Based Algorithm Development and Validation.” In:

- Sensors (Basel)* (2020). PMID: 33096899; PMCID: PMC7590046. DOI: 10.3390/s20205963.
- [16] Kimberly Kontson et al. *Wearables for gait in Parkinson's Disease and age-matched controls*. 2024. DOI: 10.7303/SYN52540892. URL: <https://repro-prod.prod.sagebase.org/repo/v1/doi/locate?id=syn52540892&type=ENTITY>.
 - [17] Gary Weiss. *WISDM Smartphone and Smartwatch Activity and Biometrics Dataset*. UCI Machine Learning Repository. DOI: <https://doi.org/10.24432/C5HK59>. 2019.
 - [18] Barbara Bruno, Fulvio Mastrogiovanni, and Antonio Sgorbissa. *Dataset for ADL Recognition with Wrist-worn Accelerometer*. UCI Machine Learning Repository. DOI: <https://doi.org/10.24432/C5PC99>. 2012.
 - [19] Carlos Polvorinos-Fernández et al. *Smartwatch dataset on six free-living activities*. Zenodo, Jan. 2025. DOI: 10.5281/zenodo.14623732. URL: <https://doi.org/10.5281/zenodo.14623732>.
 - [20] 2M Engineering Ltd. *QSense Motion Platform User Manual (QSM-210)*. Version 1.6. Sept. 2024. URL: <https://qsense-motion.com/downloads/User-Manual-QSense.pdf>.
 - [21] Dimitrios Megaritis. *Multimobility_{wrist} : Algorithms for Digital Mobility Assessment from Wrist-Worn Sensors*. 2025. DOI: 10.5281/zenodo.16926413. URL: <https://doi.org/10.5281/zenodo.16926413>.
 - [22] Jan Christian Brønd, Lars Bo Andersen, and Daniel Arvidsson. “Generating ActiGraph Counts from Raw Acceleration Recorded by an Alternative Monitor”. In: *Medicine and science in sports and exercise* (2017). PMID: 28604558. DOI: 10.1249/MSS.0000000000001344.
 - [23] Shing Chan et al. “CAPTURE-24: A large dataset of wrist-worn activity tracker data collected in the wild for human activity recognition”. In: *Scientific Data* (2024).
 - [24] Aiden Doherty et al. “Large scale population assessment of physical activity using wrist worn accelerometers: The UK Biobank study”. In: *PLOS ONE* 12.2 (2017), e0169649. DOI: 10.1371/journal.pone.0169649.
 - [25] Luca Palmerini et al. “Mobility recorded by wearable devices and gold standards: the Mobilise-D procedure for data standardization”. In: *Scientific Data* (2023). DOI: 10.1038/s41597-023-01930-9.

9 Appendix

9.1 Public wrist-worn IMU Datasets

- **CAPTURE-24:** Activity tracker dataset for human activity recognition [23]. This dataset claims to be the largest one available at the time of the realisation, with (healthy) 151 participants, amounting to 3883 hours of accelerometer data, of which 2562 hours are annotated. The participants were asked to wear the device in daily living for a period of roughly 24 hours. The annotations were made with wearable cameras and sleep diaries. It was excluded due to computational difficulties when sharing and processing the dataset (as a result of the large amount of data).
- **UK Biobank:** over 100,000 participants aged 40–69 years wearing the sensors for around 7 days [24]. The sensors worn on the dominant hand during normal daily activities. The data is not directly annotated. This dataset is available upon request.
- **Mobilize-D:** the main study focus on the lower back position of the sensors, however for the In TVS and YAR studies, additional sensors were included as INDIP (independent reference system) on the non-dominant wrist [25]. TVS data is based on 120 participants (healthy older adults, Parkinson’s Disease (PD), chronic obstructive pulmonary disease (COPD), multiple sclerosis (MS), proximal femoral fracture (PFF), and chronic heart failure (CHF). It was excluded due to difficulties of finding a structured description of the subsets in the dataset.

9.2 Signal Processing Algorithm Flowcharts

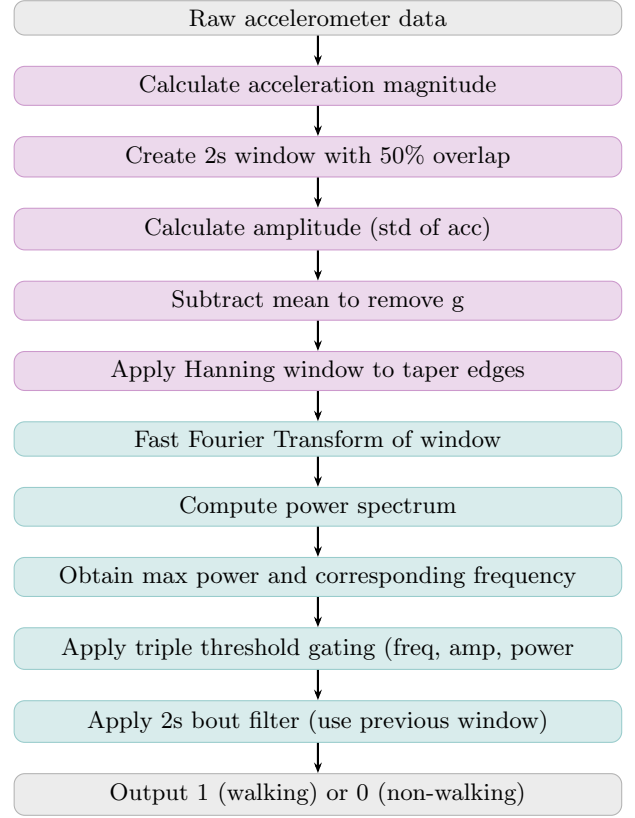


Figure 4: Signal processing pipeline of the (adapted) Straczekiewicz A algorithm. Purple nodes indicate windowing and preprocessing steps; teal nodes indicate spectral analysis and classification steps.

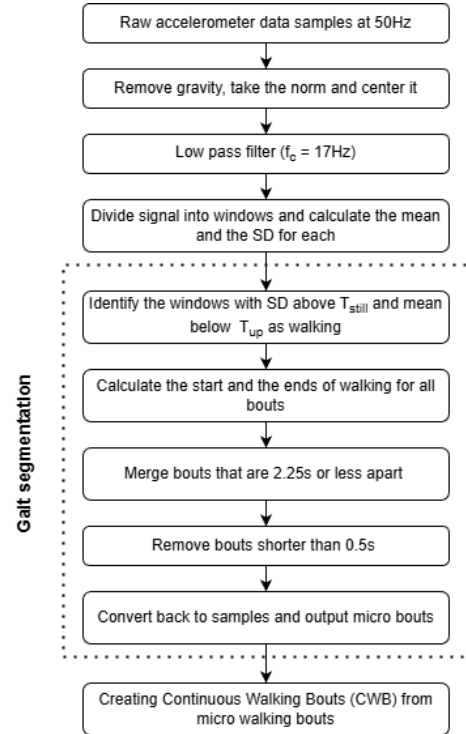


Figure 5: The flow chart showing the process of gait detection in the Hickey adaptation.

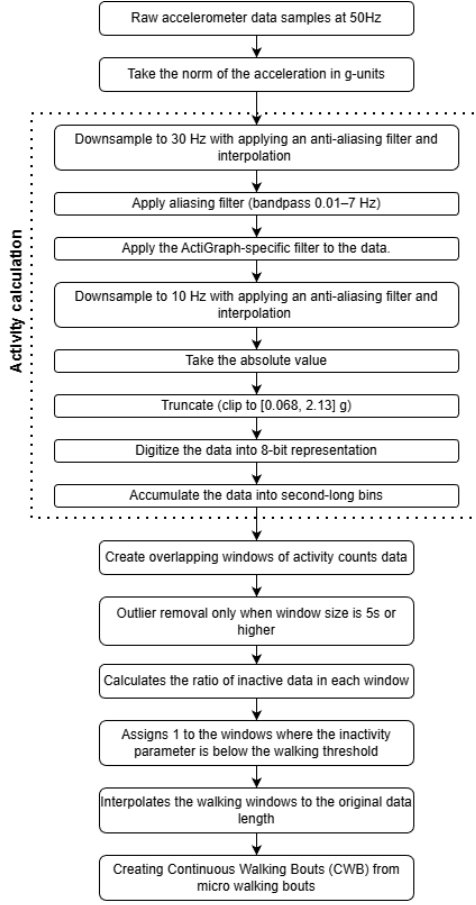


Figure 6: The flow chart showing the process of gait detection in the Kheirhahan adaptation.

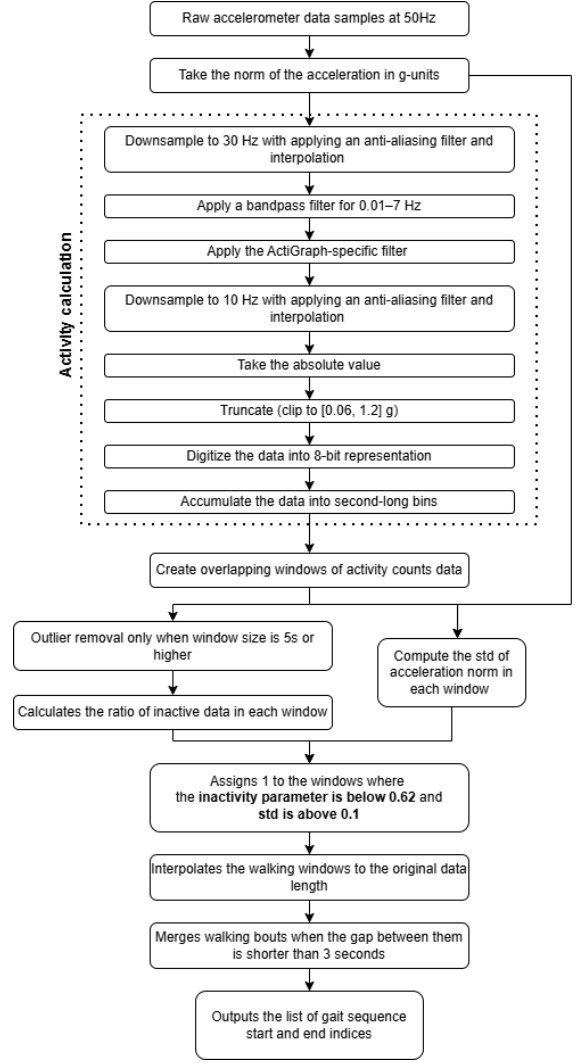


Figure 7: The flow chart showing the process of gait detection in the Kheirhahan adaptation modified with a standard threshold check.

9.3 Per-activity Breakdown of Cross-dataset analysis

Table 7: WISDM — Per-activity performance (pooled across subjects). *Jog*, *Stairs* and *Walk* are considered gait. *Chips*, *Pasta*, *Sandwich*, and *Soup* refer to eating those foods.

Activity	Strackiewicz			ElderNet			StrokeNet			Kheirkahan			Kheirkahan mod.			Realtime		
	P	R	Acc	P	R	F1	P	R	F1	P	R	Acc	P	R	F1	P	R	Acc
Catch	—	—	.358	—	—	.994	—	—	.117	—	—	.174	—	—	.177	—	—	.161
Chips	—	—	.994	—	—	.938	—	—	.458	—	—	.969	—	—	.973	—	—	.961
Clapping	—	—	.625	—	—	.908	—	—	.147	—	—	.496	—	—	.496	—	—	.405
Dribbling	—	—	.218	—	—	.965	—	—	.054	—	—	.140	—	—	.143	—	—	.130
Drink	—	—	.996	—	—	.917	—	—	.565	—	—	.947	—	—	.965	—	—	.960
Folding	—	—	.596	—	—	.995	—	—	.171	—	—	.299	—	—	.300	—	—	.233
Jog	1.	.867	.929	.867	1.	.037	.071	.037	1.	1.	1.	.934	.966	.934	1.	.994	.997	.994
Kicking	—	—	.597	—	—	.995	—	—	.127	—	—	.264	—	—	.264	—	—	.188
Pasta	—	—	.992	—	—	.936	—	—	.409	—	—	.974	—	—	.977	—	—	.971
Sandwich	—	—	.972	—	—	.929	—	—	.525	—	—	.908	—	—	.918	—	—	.907
Sit	—	—	.989	—	—	.990	—	—	.833	—	—	.972	—	—	.973	—	—	.962
Soup	—	—	.996	—	—	.956	—	—	.574	—	—	.945	—	—	.962	—	—	.953
Stairs	1.	.863	.926	.863	1.	.157	.271	.157	1.	.980	.990	.980	.945	.972	.945	1.	.983	.992
Stand	—	—	.993	—	—	.998	—	—	.845	—	—	.990	—	—	.991	—	—	.988
Teeth	—	—	.824	—	—	.992	—	—	.211	—	—	.978	—	—	.978	—	—	.951
Type	—	—	.998	—	—	.993	—	—	.592	—	—	.999	—	—	.999	—	—	.997
Walk	1.	.928	.963	.928	1.	.062	.116	.062	1.	.993	.997	.993	1.	.908	.952	.908	1.	.944
Writing	—	—	.993	—	—	.996	—	—	.644	—	—	.974	—	—	.975	—	—	.969

Table 8: BIOCLITE — Per-activity performance (pooled across subjects). *Walking 50m* considered as gait.

Activity	Strackiewicz			ElderNet			StrokeNet			Hickey			Kheirkahan			Kheirkahan mod.			Realtime		
	P	R	Acc	P	R	F1	P	R	F1	P	R	Acc	P	R	F1	P	R	Acc	P	R	Acc
Stirring a mixture	—	—	.431	—	—	.726	—	—	.160	—	—	.044	—	—	.881	—	—	.881	—	—	.846
Brushing teeth	—	—	.460	—	—	.881	—	—	.058	—	—	0.027	—	—	.660	—	—	.660	—	—	.588
Drawing spiral	—	—	.998	—	—	.906	—	—	.892	—	—	.911	—	—	.993	—	—	.993	—	—	.993
Resting (chair)	—	—	.988	—	—	.965	—	—	.951	—	—	.948	—	—	.971	—	—	.971	—	—	.960
Transitions	—	—	.605	—	—	.798	—	—	.427	—	—	.233	—	—	.592	—	—	.592	—	—	.547
Typing	—	—	.922	—	—	.629	—	—	.639	—	—	.598	—	—	.986	—	—	.986	—	—	.982
Walking 50m	1.	.947	.973	.947	1.	.392	.563	.392	1.	.999	1.	.999	1.	1.	1.	1.	1.	1.	1.	1.	1.

9.4 Raw Acceleration Data from Sub1 & Sub3 in *Clinical*

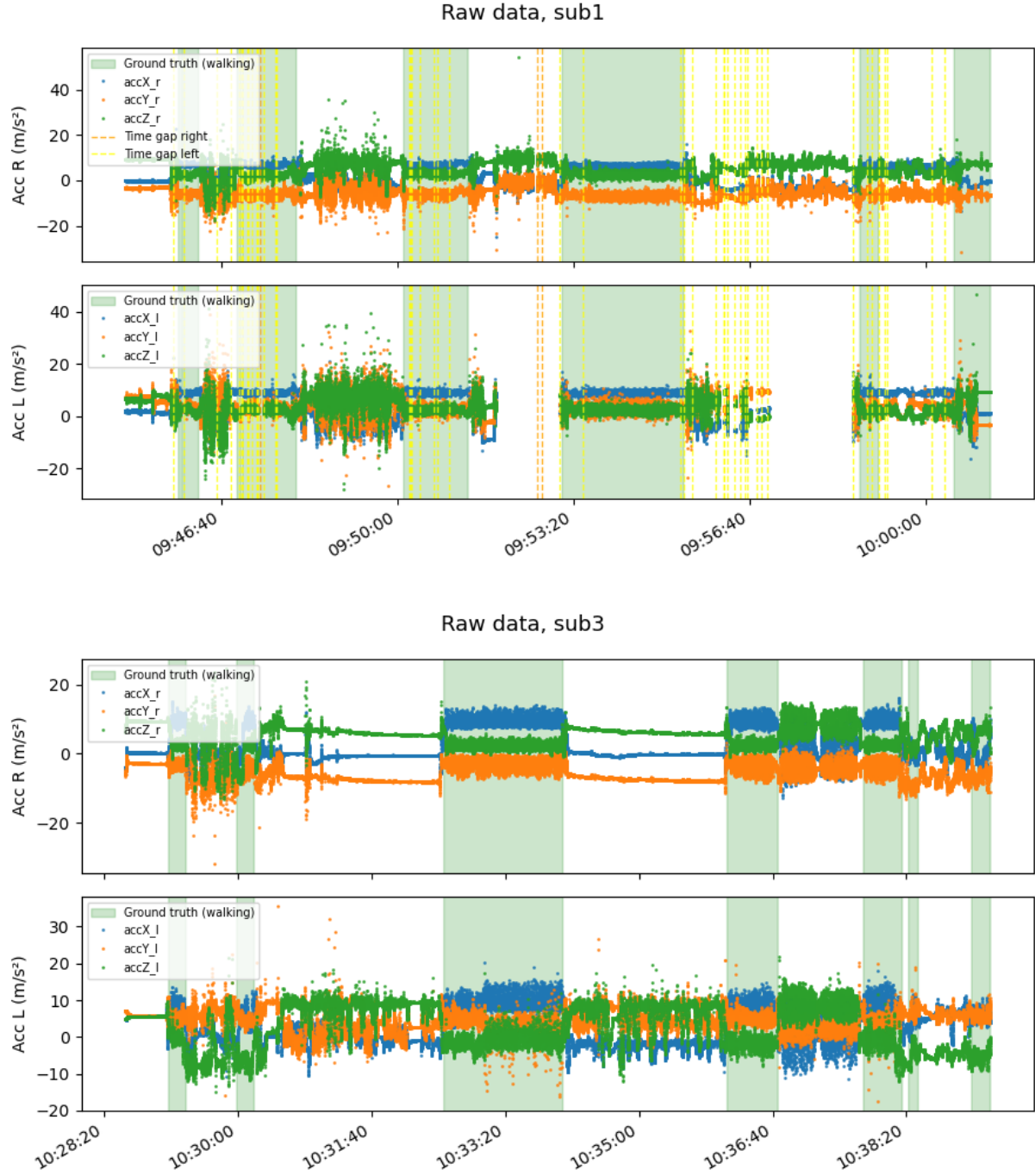


Figure 8: Raw accelerometer data from Subjects 1 (walker used) & 3 (no walker used) in *Clinical* dataset, showing the effect of the presence of the aid on the signal variability.